

DAIR³ - Unit 7.1

The Ethics of AI Agents

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©May 18, 2026



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Two kinds of claims about the world

We can weigh a cup.

The result is reproducible: anyone with a scale arrives at the same number, regardless of who they are or what they believe.

Knowledge is a *reproducible* assessment of a state of the world.

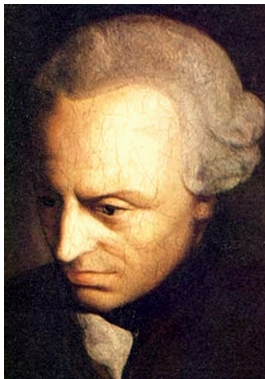
We cannot weigh a deity.

One may *opt* to believe (or not). The position is shared, adopted, or opposed — but it is not settled by reproducible assessment.

Belief is a position adopted, not proven or disproven by measurement.

The line between knowledge and belief is *reproducibility*, not subject matter.

Kant's two questions, in today's terms



Immanuel Kant (1724–1804).

Portrait c. 1790, unknown artist.

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Critique of Pure Reason (1781)

“What can we know?” — the limits of empirical reason. The cup is here.

Critique of Practical Reason (1788)

“What ought we do?” — moral law, irreducible to facts.

The two domains are categorically distinct. Facts alone cannot generate “shoulds.” Moving from *is* to *ought* requires a value layer that no measurement supplies.

Ethics is the value system we adopt

- **Ethics** is the set of beliefs about what the world *should be*. These beliefs are **values**.
- Values are **positional**: they depend on *where* and *when* one stands. There is no view from nowhere.
- **No values framework has universal warrant**. All are positionally grounded.
- **Regulation** is the mechanism by which positional values become *binding* — rules to which subjects must adhere.

When we ask “*what are the ethics of AI agents?*” we are asking **which positional values we choose to encode and enforce**. The rest of this unit is about that choice.

Discursive Networks: many humans, many machines

A **discursive network** is a large-scale ecosystem of human and machine agents whose utterances circulate, reinforce, and mutate through repeated exchange.

Machines participate in this exchange through large language models (**LLMs**), which are large artificial neural networks capable of producing linguistic outputs essentially indistinguishable from human discourse.



©2025 Juan B. Gutiérrez. Generation: GPT API

Why a network framing? Three propagation properties

Statements in a discursive network exhibit three properties that make “is this output correct?” an *insufficient* reliability question:

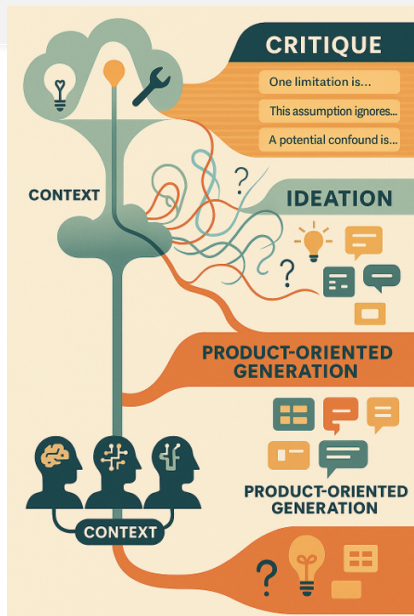
- **Circulation.** An agent’s output propagates through documents, code, presentations, and the published literature. Once in circulation, statements acquire lives of their own.
- **Authority accumulation.** Statements gain credibility through association with trusted sources, repetition, and integration into larger narratives — *independent of their origin*.
- **Mutation.** As statements move, they are paraphrased, summarized, recombined, and stripped of context.

Even when most outputs are accurate, network dynamics make errors propagate widely and hard to retract.

Engaging LLMs

LLMs support a range of functional output types that can be systematically grouped into three core categories:

- **ideation** (constructive synthesis),
- **critique** (diagnostic evaluation), and
- **product-oriented generation** (goal-directed deliverables).



What makes an AI agent different from a chatbot

A chatbot replies. An **agent** acts. Four capabilities expand the ethical surface:

- **Planning** — decomposes a goal into steps without supervision.
- **Tool use** — invokes external systems (databases, code execution, APIs).
- **Persistent memory** — carries state across sessions.
- **Multi-agent coordination** — agents call other agents.

Each capability adds a distinct failure mode. A statement produced by an agent and merged into a manuscript acquires authority *independent of its accuracy*.

Verification is therefore a *structural* responsibility, not an optional quality check.

From hallucination to invalidation



LLMs are prone to generating false or misleading statements, a phenomenon often referred to as “**hallucinations**”.

The metaphor of hallucination is limited: it implies a private sensory distortion, whereas an unfounded LLM assertion can be circulated, cited, and acted upon as fact.

This invention makes people seem omniscient

“This discovery of yours will create forgetfulness in the learners’ souls. . . you give your disciples not truth, but only the semblance of truth; they will be heroes of many things, and will have learned nothing; they will appear to be omniscient and will generally know nothing.”

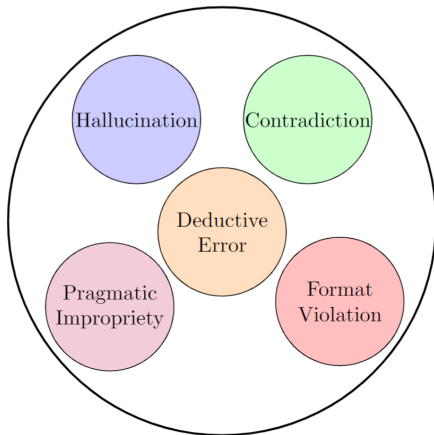
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PLATO, PHAEDRUS 274c–275b

From hallucination to invalidation

Invalidation = any output violating constraints of facts, logic, norms, or formats.



- **Hallucination:** content introduced without grounding
- **Contradiction:** statements that conflict
- **Deductive error:** conclusion failing to follow from premises
- **Pragmatic impropriety:** output violating norms
- **Format violation:** output breaking format constraints

Invalidation in LLMs: the working taxonomy

- LLMs generate human-like text but produce **invalidations**: false or misleading statements. The traditional term “hallucination” is too narrow.
- **Invalidation** encompasses: factual errors, logical contradictions, format violations, ethical breaches.
- These errors propagate through interconnected communication systems.

Four invalidation types in vital-statistics queries

Type	NCHS example	Verification strategy
Factual	LLM names a variable not present in the DVS schema as a named field.	Schema lookup in dictionary.
Logical	LLM recommends a rate denominator inconsistent with the research-question population.	Re-derive denominator from the question.
Normative	LLM interprets racial birthweight disparities as biological differences.	Sociodemographic-context injection (block 3).
Structural	LLM queries APGAR columns for records before 1977 (columns did not exist).	Year-banded dictionary cross-check.

Every invalidation type requires a *type-specific* verification strategy. A generic “double-check it” does not work. When a failure could be classified more than one way, classify it by the constraint that fails *first* in the appropriate verification workflow.

One workflow, four invalidation types

Literature triage for a Texas county-level underweight-birth study

- **Factual.** Retrieval agent cited a 2019 paper that does not exist.
Verification: every DOI checked against the publisher's resolver before retention. Bibliography re-audited; one further fabrication found.
- **Logical.** Draft claimed an underweight-rate difference between counties after citing an odds ratio — but the OR was for the maternal-age covariate, not the county effect.
Verification: every statistical claim traced to its source table; rewrite.
- **Normative.** Draft described racial disparities in birth outcomes as “intrinsic to the population.”
Verification: human author replaced framing with social-determinants language consistent with the published literature.
- **Structural.** CSV summary's column order did not match the visualization tool's schema.
Verification: schema validation failed at the boundary; agent file.

One workflow surfaced all four invalidation types. Each required a different verification.

Key questions

- How do we characterize mathematically a discursive network?
- How do invalidations propagate in a discursive network?
- Are there mathematically-driven strategies to minimize the presence of invalidations in the final state of a discursive exchange?

Answers in **Unit 7.2**. This unit (7.1) establishes the ethical framing those answers require.

Sociodemographic variables: social constructs with real consequences

Race has no valid biological basis. Its reification in law, medicine, and data systems makes it *real in its consequences*.

Birth outcomes differ by recorded race not because of intrinsic biology, but because of **racism operating through identifiable pathways**:

- Differential access to prenatal care
- Chronic stress from discrimination
- Environmental exposures in historically redlined neighborhoods
- Differential clinical treatment

An LLM that reports a racial disparity without naming the pathway commits a *normative* invalidation.

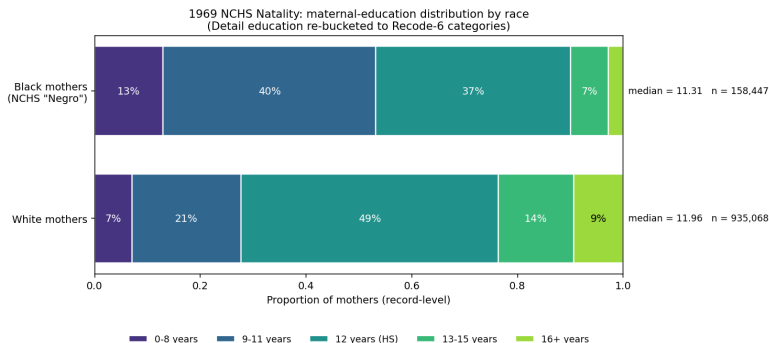
Race in the NCHS 1969–1986 data: an unstable variable

- **1969–1977.** Race assigned from a fixed set: {White, Black, American Indian, Chinese, Japanese, Other}. For parents of different recorded races, the *child's race follows the mother's*.
- **1978.** Assignment rules revised. The revision propagated *inconsistently across states*.
- **Hispanic ethnicity.** Category appeared late in the study period. Collapsed into “White” in some years.
- **child_race_recode_3.** The field's encoding changed midway through the study period.

Implication. Any LLM query about race spanning 1969–1986 risks *normative* invalidation (treating categories as stable) and *structural* invalidation (treating encoding as stable).

The same pattern holds for maternal education and geographic identifiers.

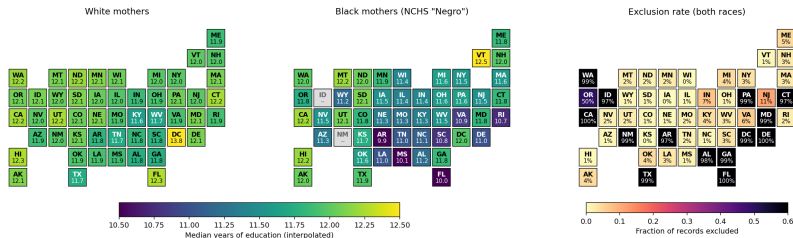
What 1969 data show — and what they encode



The distributions differ. Interpolated medians: White ≈ 11.96 , Black ≈ 11.31 years of education. The gap is real *and* it is **structural, not biological**: segregated schools, redlined neighborhoods, and school funding tied to local property tax produced this distribution. To read the data is one thing; to locate the cause is another.

Within-state pattern — and the missingness

1969 NCHS Natality: maternal education by state of residence



Left and middle panels: median maternal education, by state of residence. Both medians rise and fall *together* by state — the gap is consistent, the level is geographic. **Right panel:** 38–41% of records are excluded because the mother's *occurrence* state did not report education in 1969. The missingness is itself a state-policy artifact; the data encodes which states deprioritized recording maternal education at all.

Seven risk categories in agentic biomedical research

- 1 **Epistemic.** What does the agent know, and how do you know?
- 2 **Accountability & authorship.** Who owns the output?
- 3 **Provenance & reproducibility.** Can the result be re-derived?
- 4 **Confidentiality.** Where does the prompt land?
- 5 **Bias & fairness.** Whose data trained this?
- 6 **Security.** What can the agent execute?
- 7 **Sustainability.** What does this cost the planet?

Each category maps to a specific clause in the governance protocol.

Multi-agent critique: what you gain and what you give up

- **Gain:** cross-detection reduces invalidation rate (quantified in Unit 7.2).

Costs to manage:

- **Privacy amplification.** The prompt now reaches n vendors, not one.
- **Epistemic homogenization.** Critics from the same family share blind spots.
- **Compute burden.** n agents = $n \times$ energy.
- **Attribution diffusion.** “The ensemble decided” is not an accountable subject.

The governance protocol must name how each cost is managed in your team's workflow.

Ethical concerns inherent to agentic systems

Epithesis: Claiming authorship without genuine contribution.

Energy costs: $O(N^2 \log N)$ computational complexity in cross-checking ensembles.

Epistemic diversity: Risk of homogenization through consensus.

Privacy: Cross-agent visibility increases the attack surface.

Mitigation strategies:

- Cryptographic integrity verification of agent contributions
- Governance frameworks for fair access
- Diversity-preserving consensus mechanisms
- Privacy-preserving critique protocols

Cultural implications

We have entered a new cultural mode:

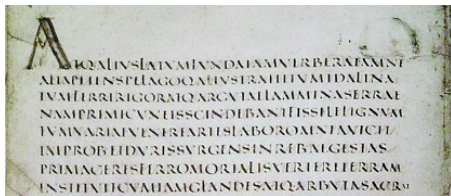
- Writing is now **co-constructed** with machines.
- Authorship becomes **distributed** across agents.
- Traditional metrics (originality, citations) need updating.
- The medium itself **acts on the message**.

Open research directions:

- ① Cultural assimilation and ethical frameworks.
- ② Redesign of quality metrics for co-creative media.
- ③ Governance of distributed agency.

We are learning a new medium

It took millennia to learn the old medium (cf. McLuhan's "*the medium is the message*," 1964). In discursive networks, **the medium acts on the message**.



Scriptum Continua. MediaWiki. Public domain. Accessed

June 2025. Vergilius_Augustus,_Georgica_141.jpg

- Space between words: 7th–8th century.
- Table of contents: Pliny's Natural History (77 CE).
- Punctuation: 17th century.
- *et cetera*.

We are just learning how to interact with LLMs. This landscape will continue to evolve. Treat this talk (and all current knowledge) as preliminary.

Governance-by-design: five core principles

Responsible agent use is *proactive design*, not reactive correction.

- ➊ **Human accountability is non-delegable.** Researchers bear full responsibility regardless of agent contributions. Agents are tools, not collaborators in the accountability sense.
- ➋ **Transparency scales with stakes.** Routine internal work needs less formality than peer-reviewed publications or clinical applications.
- ➌ **Verification is part of the workflow.** Not a finishing polish; an explicit stage appropriate to the output type.
- ➍ **Privacy protection is a precondition.** Sensitive data does not enter agent workflows without safeguards. *When in doubt, exclude.*
- ➎ **Reproducibility requires records.** A workflow that cannot be reconstructed from available logs has not met the standard expected of scientific research.

Practical checklist before deploying agents

- ❶ **Scope.** What can agents decide?
What requires human approval?
- ❷ **Data.** PHI? Unpublished?
Confidential peer-review material?
Is the system approved for the
data class?
- ❸ **Provenance.** Are prompts,
outputs, tool calls, and model
versions logged retrievably?
- ❹ **Verification.** Which outputs
require independent verification,
by whom, by what method?
- ❺ **Authorship.** Who meets
authorship criteria? How is AI
assistance disclosed?
- ❻ **Bias.** Have outputs been
evaluated for biased framing and
group harms?
- ❼ **Security.** Are tools sandboxed?
Are permissions minimized? Is
retrieved content screened?
- ❽ **Sustainability.** Is the verification
overhead proportionate to the risk
level?

Research AI Governance Protocol: five components

Every team produces a one-page protocol with:

- ➊ **Provenance logging.** How prompts, model IDs, and timestamps are recorded for every ensemble interaction.
- ➋ **Invalidation log format.** The template for recording invalidation type, description, and resolution.
- ➌ **Sociodemographic handling clause.** How variables with unstable encoding are queried; how normative claims are flagged.
- ➍ **Escalation threshold.** What output is *not* used without independent verification.
- ➎ **Draft disclosure statement.** Tools, tasks, and accountability — finalized at the end of Unit 7.3.

The protocol is a **live document** — updated through Units 7.2 and 7.3 as invalidations are encountered.

Deliverables for Unit 7.1

Item 1 — Invalidation Taxonomy Application (in-session, 8 min)

- Four instructor-presented NCHS scenarios.
- For each: (a) invalidation type, (b) why this type not another, (c) appropriate verification method.

Item 2 — Research AI Governance Protocol (in-session construction, 10 min)

- One-page document, five required components (previous slide).
- Active from the start of Unit 7.2.
- Disclosure statement finalized at end of Unit 7.3 (Item 9).



Thanks for your attention!

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